Forces And Motion

STARTER

1. Calculate $(2\mathbf{i}+\mathbf{j})+(3\mathbf{i}-4\mathbf{j})$ and $(-\mathbf{i}+3\mathbf{j})-(3\mathbf{i}-\mathbf{j})$.
2. A right triangle has perpendicular sides 12 cm and 15 cm. Find its hypotenuse and the angle opposite the 12 cm side to 1 d.p.
3. A car accelerates from rest at $1.5\ \text{m/s}^2$ for 12 s. Find its speed. It then decelerates at $1\ \text{m/s}^2$: find its stopping distance.

Solution 2

1

1. $5\mathbf{i}-3\mathbf{j}$; $-4\mathbf{i}+4\mathbf{j}$.

2

2. $\sqrt{12^2+15^2}=19.2$ cm;
$\tan^{-1}(12/15)=38.7°$.

3

3. $v=1.5(12)=18$ m/s.
$0=18^2-2s\Rightarrow s=162$ m.

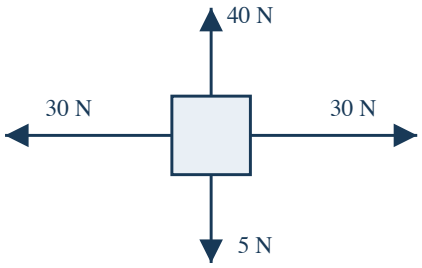
Force diagrams

Weight $W = mg$ acts vertically downwards. Normal reaction R is perpendicular to the contact surface. Tension acts along a taut string, away from the body. Friction opposes relative sliding or its tendency.

Newton's first law

Zero resultant force means constant velocity (including rest).

An object has forces 30 N right, 30 N left, 40 N up and 5 N down. Find the resultant.



$F_x = 30 - 30 = 0, \qquad F_y = 40 - 5 = 35.$

The resultant is 35 N upwards. Acceleration is in the direction of the resultant force.

1. Forces act on a particle: P N right, 3 N left, 5 N up and 5 N down. Find the resultant.
2. Forces act on another particle: 5 N right, 7 N left, 4 N up and 6 N down. Find the resultant and its magnitude.

1

1. $\mathbf{F} = (P - 3)\mathbf{i}$ N. Its direction depends on whether P is greater or less than 3.

2

2. $\mathbf{F} = -2\mathbf{i} - 2\mathbf{j}$ N.

   $|\mathbf{F}| = \sqrt{8} = 2\sqrt{2}$ N.

   It points diagonally down and left.

Forces as Vectors

Solution 5

1. Forces $(2,3),(4,-1),(-3,2),(a,b)$ N are in equilibrium. Find a,b.
2. With $\mathbf{i}$ east and $\mathbf{j}$ north, find the resultant, magnitude and bearing of forces $(2,1),(3,-2),(-1,4)$ N.

1

1. $(3+a,4+b)=(0,0)$, so $a=-3, b=-4$.

2

2. $\mathbf{R}=4\mathbf{i}+3\mathbf{j}$ N.
   $|\mathbf{R}|=5$ N.
   $\theta=\tan^{-1}(4/3)=053.1°$.

Test Your Understanding

Solution 6

1. Forces $(7\mathbf{i}-9\mathbf{j}),(5\mathbf{i}+6\mathbf{j}),(p\mathbf{i}+q\mathbf{j})$ N are in equilibrium. Find p,q. Remove the third force: find the resultant's magnitude and angle with $\mathbf{j}$, to the nearest degree.
2. Forces $(\mathbf{i}-3\mathbf{j})$ N and $(p\mathbf{i}+2p\mathbf{j})$ N act on a particle. Find the first force's angle with $\mathbf{j}$, and $p>0$ if the resultant is parallel to $\mathbf{i}$.

1

1. $p=-12, q=3$.
   $\mathbf{R}=12\mathbf{i}-3\mathbf{j}$.
   $|\mathbf{R}|=3\sqrt{17}=12.4$ N.
   Angle with $\mathbf{j}$:
   $90°+\tan^{-1}(3/12)=104°$.

2

2. Angle
   $=180°-\tan^{-1}(1/3)=161.6°$.
   $-3+2p=0\Rightarrow p=\frac{3}{2}$.

Exercise 10B – Y1 Mech Pages 161-162

3. Forces $(a, 2b), (-2a, -b), (3, -4)$ N act on an object in equilibrium. Find a, b.

7. Forces $(2a\mathbf{i} + 2b\mathbf{j}), (-5b\mathbf{i} + 3a\mathbf{j}), (-11\mathbf{i} - 7\mathbf{j})$ N are in equilibrium. Find a, b.

Solution 7

3. $-a + 3 = 0$ and $b - 4 = 0$.

$a = 3, \quad b = 4.$

7. $2a - 5b = 11, 3a + 2b = 7.$

Eliminate b: $19a = 57$.

$a = 3, \quad b = -1.$

Exercise 10B – Y1 Mech Pages 161-162

9. Forces $\mathbf{F}_1 = (3\mathbf{i} - 2\mathbf{j})$ N and $\mathbf{F}_2 = (a\mathbf{i} + 2a\mathbf{j})$ N act on a particle, where $a > 0$.

a. Find the angle between $\mathbf{F}_2$ and $\mathbf{i}$.

b. Their resultant is parallel to $13\mathbf{i} + 10\mathbf{j}$. Find a.

Solution 8

a

a. $\theta = \tan^{-1}(2) = 63.4°$.

b

b. $\mathbf{R} = (3 + a)\mathbf{i} + (2a - 2)\mathbf{j}$.

$13(2a - 2) = 10(3 + a)$

$16a = 56$

$a = 3.5$.

10. Three forces are $\mathbf{F}_1 = (-7, -4)$ N, $\mathbf{F}_2 = (4, 2)$ N and $\mathbf{F}_3 = (a, b)$ N.

a. Given equilibrium, find a, b.

b. Remove $\mathbf{F}_1$. Find the resultant's magnitude and angle above the horizontal, to the nearest degree.

a

a. $-7 + 4 + a = 0$, $-4 + 2 + b = 0$.

$a = 3, \quad b = 2$.

b

b. $\mathbf{R} = (7, 4)$ N.

$|\mathbf{R}| = \sqrt{65} = 8.06$ N.

$\theta = \tan^{-1}(4/7) = 30°$.

Extension

A force F_1 acts in the direction of $\mathbf{i}$; a force F_2 acts at angle θ to $\mathbf{i}$. Show that the resultant has magnitude

$$\sqrt{F_1^2 + F_2^2 + 2F_1F_2\cos\theta}.$$

$\mathbf{R} = (F_1 + F_2\cos\theta)\mathbf{i} + F_2\sin\theta\mathbf{j}$

$|\mathbf{R}|^2 = (F_1 + F_2\cos\theta)^2 + F_2^2\sin^2\theta$

$= F_1^2 + 2F_1F_2\cos\theta + F_2^2(\cos^2\theta + \sin^2\theta)$

$= F_1^2 + F_2^2 + 2F_1F_2\cos\theta.$

Take the non-negative square root.

Forces and Acceleration

Solution 11

Newton's second law

$\mathbf{F}_{\text{resultant}} = m\mathbf{a}$

1. A 2000 kg car has driving force 800 N and resistance 200 N. Find its acceleration.
2. Find the weight of a 50 kg child.
3. A falling 70 kg body experiences 300 N air resistance. Find its acceleration.

1

1. $800 - 200 = 2000a$.

$a = 0.3$ m/s^2.

2

2. $W = 50(9.8) = 490$ N downwards.

3

3. Downwards positive:

$70g - 300 = 70a \Rightarrow a = 5.51$ m/s^2.

Combining F=ma with suvat equations

Solution 12

A 5 kg body is pulled from rest along a rough horizontal table by a horizontal force of 20 N. Friction is 4 N. Find:

a. its acceleration;
b. its displacement in the first 4 s;
c. the normal reaction.

a

a. $20 - 4 = 5a \Rightarrow a = 3.2$ m/s^2.

b

b. $s = \frac{1}{2}(3.2)(4^2) = 25.6$ m.

c

c. No vertical acceleration:

$R - 5g = 0 \Rightarrow R = 49$ N.

Testing Your Understanding

Solution 13

A particle of mass 4 kg reaches soft horizontal ground with downward speed 28 m/s. The ground exerts a constant upward resistive force of 5000 N. Find how far the particle sinks before coming to rest.

Downwards positive:

$4g - 5000 = 4a \Rightarrow a = -1240.2\ \text{m/s}^2$.

$0 = 28^2 + 2(-1240.2)s$

$s = 0.316\ \text{m}\ (3\ \text{s.f.})$.

Exercise 10C – Y1 Mech Pages 165

Solution 14

9. A lift of mass 500 kg carries passengers of total mass 300 kg. It accelerates from rest to 3 m/s over 5 m. Find:

a. the acceleration magnitude;

b. cable tension when accelerating downwards;

c. cable tension when accelerating upwards.

a

a. $3^2 = 2a(5) \Rightarrow a = 0.9\ \text{m/s}^2$.

b

b. $800g - T = 800(0.9) \Rightarrow T = 7120$ N.

c

c. $T - 800g = 800(0.9) \Rightarrow T = 8560$ N.

Exercise 10C – Y1 Mech Pages 165

10. A 50 kg trolley is pulled from rest along a horizontal path by a horizontal rope. After 2 s its speed is 1 m/s. Constant resistance is 20 N. Find:

a. its acceleration;

b. rope tension.

Solution 15

a

a. $a = (1 - 0)/2 = 0.5\ \text{m/s}^2$.

b

b. $T - 20 = 50(0.5) \Rightarrow T = 45\ \text{N}$.

Exercise 10C – Y1 Mech Pages 165

11. A 400 kg van moves at 16 m/s when its engine cuts out. It stops without braking. Model the resistance as a constant 200 N. Find:

a. the stopping time;

b. stopping distance;

c. a limitation of this model.

Solution 16

a

a. $-200 = 400a \Rightarrow a = -0.5\ \text{m/s}^2$.

$0 = 16 - 0.5t \Rightarrow t = 32\ \text{s}$.

b

b. $0 = 16^2 + 2(-0.5)s \Rightarrow s = 256\ \text{m}$.

c

c. Air resistance usually depends on speed, so constant resistance is an approximation.

Motion in two dimensions

Force, acceleration, velocity and displacement are vectors. Mass, area and volume are scalars.

A resultant force $(3\mathbf{i} + 8\mathbf{j})$ N acts on a 0.5 kg particle. Take **i** east and **j** north. Find its acceleration vector, magnitude and bearing.

Solution 17

$\mathbf{a} = \dfrac{\mathbf{F}}{m} = 6\mathbf{i} + 16\mathbf{j}\ \text{m/s}^2.$

$|\mathbf{a}| = \sqrt{292} = 17.1\ \text{m/s}^2$

$\theta = \tan^{-1}(6/16) = 020.6°.$

A 60 kg boat experiences forces $(80, 50)$, $(10p, 20q)$ and $(-75, 100)$ N. Its acceleration is $(0.8, -1.5)$ m/s^2. Find p, q.

Solution 18

$(5 + 10p, 150 + 20q) = (48, -90).$

$5 + 10p = 48 \Rightarrow p = 4.3$

$150 + 20q = -90 \Rightarrow q = -12.$

3. A particle of mass 3 kg has acceleration $(7\mathbf{i} - 3\mathbf{j})$ m/s^2. Find its resultant force, magnitude and bearing. Take **i** east and **j** north.

3
$\mathbf{F} = 3(7\mathbf{i} - 3\mathbf{j}) = 21\mathbf{i} - 9\mathbf{j}$ N.

3
$|\mathbf{F}| = \sqrt{522} = 22.8$ N
Bearing $= 90^\circ + \tan^{-1}(9/21) = 113^\circ$
to the nearest degree.

6. Forces $(2\mathbf{i} + 3\mathbf{j})$ N and $(p\mathbf{i} + q\mathbf{j})$ N have resultant parallel to $-\mathbf{i} + 4\mathbf{j}$. Show that $4p + q + 11 = 0$.

6
Write $\mathbf{R} = k(-\mathbf{i} + 4\mathbf{j})$. Comparing components:
$2 + p = -k, \qquad 3 + q = 4k.$

6
$3 + q = 4(-2 - p) \Rightarrow 4p + q + 11 = 0.$

Exercise 10D – Y1 Mech Pages 167-169

Solution 21

7. A 4 kg particle starts from rest under constant force $(6\mathbf{i} + b\mathbf{j})$ N, bearing 045°. Find:

a. b;

b. force magnitude;

c. acceleration magnitude;

d. distance travelled in 5 s.

a

a. Equal positive components:
$b = 6$.

b. $|\mathbf{R}| = 6\sqrt{2}$ N.

c

c. $a = 6\sqrt{2}/4 = 3\sqrt{2}/2$ m/s^2.

d

d. $s = \frac{1}{2}\dfrac{3\sqrt{2}}{2}(5^2) = \dfrac{75\sqrt{2}}{4} = 26.5$ m.

Exercise 10D – Y1 Mech Pages 167-169

Solution 22

10. Forces $(2, 5)$ N and (p, q) N act on a particle of mass m.

a. Their resultant is parallel to $(1, -2)$. Show $2p + q + 9 = 0$.

b. Given $p = 1$ and acceleration magnitude $15\sqrt{5}$ m/s^2, find m.

a

a. $2 + p = k,\ 5 + q = -2k$.
$5 + q = -2(2 + p)$
$2p + q + 9 = 0$.

b

b. $q = -11$, so $\mathbf{R} = (3, -6)$ N.
$m = \dfrac{3\sqrt{5}}{15\sqrt{5}} = 0.2$ kg.

Challenge

A 0.5 kg particle experiences forces $-4\mathbf{i}$ N and $(k\mathbf{i} + 2k\mathbf{j})$ N. Its acceleration magnitude is $8\sqrt{17}$ m/s². Find $k > 0$.

Challenge

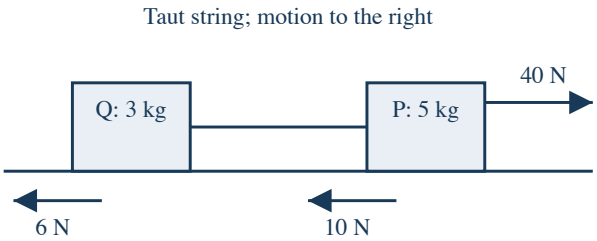
$\sqrt{(k-4)^2 + 4k^2} = 4\sqrt{17}$
$5k^2 - 8k - 256 = (k - 8)(5k + 32) = 0$.
Since $k > 0$, $k = 8$.

Connected Particles

Solution 23

Particles P, Q, masses 5 kg and 3 kg, are joined by a light inextensible string on a rough horizontal plane. A horizontal 40 N force pulls P away from Q. Friction is 10 N on P and 6 N on Q.

a. Find their acceleration.



For the whole system the internal tensions cancel:

$40 - 10 - 6 = (5 + 3)a \Rightarrow a = 3\ \text{m/s}^2$.

Alternatively, for the separate particles:

$40 - T - 10 = 5a, \qquad T - 6 = 3a.$

Adding gives $24 = 8a$, as before.

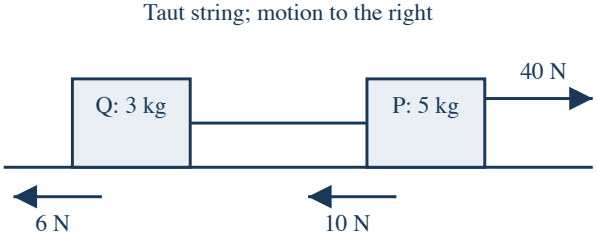
Connected Particles

Solution 24

Particles P, Q, masses 5 kg and 3 kg, are joined by a light inextensible string on a rough horizontal plane. A horizontal 40 N force pulls P away from Q. Friction is 10 N on P and 6 N on Q.

b. Find the tension.

c. Explain the string assumptions.



b

b. For Q: $T - 6 = 3(3) \Rightarrow T = 15$ N.

c

c. Light: negligible mass and uniform tension.

Inextensible and taut in this arrangement: both particles have the same acceleration.

Newton's third law

Interaction forces on two different bodies are equal in magnitude and opposite in direction.

A light scale-pan carries A (0.4 kg) on top of B (0.6 kg). A light vertical string raises the pan with acceleration 0.5 m/s^2. Find:

a. string tension;

b. force on B by A;

c. force on B by the pan.

a

a. $T - 1g = 1(0.5) \Rightarrow T = 10.3$ N.

b

b. For A: $R - 0.4g = 0.4(0.5) \Rightarrow R = 4.12$ N. Hence A exerts 4.12 N downwards on B.

c

c. For both masses: $N - 1g = 1(0.5) \Rightarrow N = 10.3$ N upwards.

A 50 kg woman stands in a 950 kg lift. It moves upwards with constant deceleration 2 m/s^2. Find:

a. cable tension;

b. the force on the woman by the floor.

a

a. Upwards positive, acceleration is -2:

$T - 1000g = 1000(-2)$

$T = 7800$ N.

b

b. $R - 50g = 50(-2) \Rightarrow R = 390$ N.

1. Particles P, Q of masses 8 kg and 2 kg are joined by a light inextensible taut string on a smooth horizontal plane. A force F pulls P away from Q, giving acceleration 0.4 m/s^2. Find F, tension and explain the assumptions.

1
$F = (8+2)(0.4) = 4$ N.

For Q: $T = 2(0.4) = 0.8$ N.

1
Light gives uniform tension and negligible string mass. Inextensible and taut gives equal accelerations along this straight string.

5. A lorry of mass m kg tows a trailer of mass $3m$ kg using a light inextensible tow-bar. Driving force is 50,000 N; resistances are 4,000 N on the lorry and 10,000 N on the trailer. Acceleration is 5 m/s^2. Find both masses and tow-bar tension; explain the assumptions.

5
$50000 - 4000 - 10000 = 4m(5)$

$m = 1800$.

Masses: 1800 kg and 5400 kg.

5
For trailer: $T - 10000 = 5400(5)$, so $T = 37000$ N.

5
The light bar has negligible mass; its end forces have equal magnitudes. Its fixed length gives equal accelerations.

6. Particles A, B of masses 10 kg and 5 kg are joined by a light inextensible vertical string, with B below A. A 180 N upward force acts on A. Find acceleration and tension.

6
$180 - 15g = 15a$

$a = 2.2\ \text{m/s}^2$ upwards.

6
For B: $T - 5g = 5(2.2) \Rightarrow T = 60$ N.

7. Particles A, B, masses 6 kg and m kg, are joined by a light inextensible vertical string, with B below A. A force of 118 N acts upwards on A, producing upward acceleration $2\ \text{m/s}^2$. Find m and the tension.

7
$118 - (6+m)g = (6+m)2$

$6 + m = 118/11.8 = 10 \Rightarrow m = 4$ kg.

7
$T - 4g = 4(2) \Rightarrow T = 47.2$ N.

8. A 6400 kg engine pulls a 1600 kg carriage using a light horizontal coupling. Driving force is 12,000 N; resistances are R N on the engine and 2000 N on the carriage. Acceleration is 0.5 m/s^2. Find R and show the coupling tension is 2800 N.

8

$12000 - R - 2000 = 8000(0.5)$

$R = 6000$ N.

8

For carriage:

$T - 2000 = 1600(0.5) \Rightarrow T = 2800$ N.

9. A 900 kg car pulls a 300 kg trailer through a light tow-bar. Resistances are 200 N and 100 N respectively; driving force is 1200 N.

a. Show acceleration is 0.75 m/s^2.

b. Find tension.

c. The engine force becomes zero and brakes exert F N on the car. Resistances stay the same and the bar's thrust is 100 N. Find F.

a

a. $1200 - 200 - 100 = 1200a$.

$a = 0.75$.

b

b. $T - 100 = 300(0.75) \Rightarrow T = 325$ N.

c

c. Trailer deceleration:

$d = (100 + 100)/300 = 2/3$.

For car, backwards positive:

$F + 200 - 100 = 900(2/3) \Rightarrow F = 500$ N.

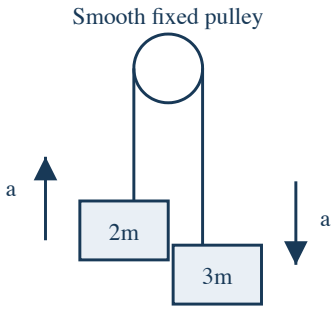
Pulleys

Solution 33

Pulleys

Particles P, Q, masses $2m, 3m$, hang from a light inextensible string over a small smooth fixed pulley. Release from rest with string taut.

Find their acceleration, tension, force on the pulley by the string, and Q's displacement after 4 s, assuming neither particle reaches an obstruction.



For P upwards and Q downwards:

$T - 2mg = 2ma, \quad 3mg - T = 3ma.$

$a = g/5, \quad T = 12mg/5.$

Both string forces on the pulley act downwards:

$F = 2T = 24mg/5.$

$s = \frac{1}{2}(g/5)(4^2) = 15.7$ m.

A light string over a smooth pulley has equal tension on both sides. The taut inextensible string gives equal acceleration magnitudes in this fixed-pulley arrangement.

Horizontal and vertical string

Solution 34

Particles A (0.4 kg) and B (0.8 kg) are connected by a light inextensible string over a smooth fixed table-edge pulley. A is on a rough table, 4.5 m from the pulley; B hangs 0.5 m above ground. Friction on A is $0.08g$ N. Released from rest, find acceleration, time until B lands, and total distance travelled by A.

$T - 0.08g = 0.4a, \quad 0.8g - T = 0.8a$

$a = 0.6g = 5.88$ m/s^2.

$0.5 = \frac{1}{2}(5.88)t^2 \Rightarrow t = 0.412$ s.

At landing, $v^2 = 2(5.88)(0.5) = 5.88$.

The string becomes slack. For A,

$a = -0.08g/0.4 = -1.96.$

$0 = 5.88 - 2(1.96)s \Rightarrow s = 1.5$ m.

Total distance $= 0.5 + 1.5 = 2.0$ m.

Test Your Understanding

Solution 35

Particles A, B, masses $5m, km$ with $k < 5$, hang at equal heights from a light inextensible string over a smooth fixed pulley. Released from rest, A descends at acceleration $g/4$.

a. Show $T = 15mg/4$.

b. Find k.

c. Explain "smooth".

d. A hits ground after 1.2 s. Find B's greatest height above ground, assuming it does not reach the pulley.

a

a. $5mg - T = 5m(g/4)$.

$T = 15mg/4$.

b. $T - kmg = km(g/4) \Rightarrow k = 3$.

c. Equal tensions on either side of the pulley.

d

d. Initial height

$h = \frac{1}{2}(g/4)(1.2^2) = 0.18g$.

When A lands, B has height $2h$ and speed $0.3g$. Additional rise:

$$s = \frac{(0.3g)^2}{2g} = 0.045g.$$

Greatest height $= 0.405g = 3.97$ m.

Exercise 10F – Y1 Mech Pages 175-177

Solution 36

1. Masses 4 kg and 3 kg hang from a light inextensible string over a smooth fixed pulley, released from rest.

a. Find tension.

b. The 4 kg mass falls 2 m before landing. Find its impact speed.

c. Find the other mass's greatest height above its initial position, assuming it does not reach the pulley.

1

$4g - T = 4a, \quad T - 3g = 3a \Rightarrow a = g/7$.

$T = 24g/7 = 33.6$ N.

1

$v^2 = 2(g/7)(2) = 5.6 \Rightarrow v = 2.37$ m/s.

1

Additional free rise $= v^2/(2g) = 2/7$ m.

Total rise $= 2 + 2/7 = 16/7$ m.

2. Masses $km, 3m$ hang initially at equal heights from a light inextensible string over a smooth fixed pulley. On release the heavier mass descends at $g/3$ and hits ground after 1.8 s.

a. Find tension.

b. Show $k = 1.5$.

c. Explain "smooth".

d. Show the lighter mass reaches height $1.26g$ m above ground without reaching the pulley.

2

$3mg - T = 3m(g/3) \Rightarrow T = 2mg.$

$T - kmg = km(g/3) \Rightarrow k = 1.5.$

Smooth pulley gives equal tensions.

2

Initial height $h = \frac{1}{2}(g/3)(1.8^2) = 0.54g.$

Speed at landing $v = 0.6g$. Extra rise $= v^2/(2g) = 0.18g.$

$H = 2(0.54g) + 0.18g = 1.26g$ m.

3. Masses $A = m$ kg and $B = 3$ kg, with $m > 3$, hang from a light inextensible string over a smooth fixed pulley. Released from rest, A descends 2.5 m in 1.25 s and stops on the ground.

a. Show acceleration is 3.2 m/s^2.

b. Find tension.

c. Show $m = 65/11$.

a

a. $2.5 = \frac{1}{2}a(1.25^2) \Rightarrow a = 3.2$ m/s^2.

b

b. $T - 3g = 3(3.2) \Rightarrow T = 39$ N.

c

c. $mg - T = ma \Rightarrow m = 39/(9.8 - 3.2) = 65/11$ kg.

3. Continued: A hits the ground after descending for 1.25 s with acceleration 3.2 m/s^2. The string becomes slack and B moves freely without reaching the pulley.

d. Explain how inextensibility was used.

e. Find the time from impact until the string becomes taut again.

d

d. While taut, the fixed string length makes the particles' acceleration magnitudes equal. Equal tension instead follows from the light string and smooth pulley.

e

e. At impact, $u_B = 3.2(1.25) = 4$ m/s. The string becomes taut when B returns to this height:

$0 = 4t - 4.9t^2 \Rightarrow t = \dfrac{40}{49} = 0.816$ s.

4. A 5 kg particle on a rough horizontal table is connected to a hanging 3 kg particle by a light inextensible string over a smooth fixed pulley. Friction is 24.5 N. Released from rest, find acceleration and tension.

4

$T - 24.5 = 5a, \qquad 3g - T = 3a$

$a = (29.4 - 24.5)/8 = 0.6125 \text{ m/s}^2.$

4

$T = 24.5 + 5(0.6125) = 27.5625$ N.

To 3 s.f.: $a = 0.613 \text{ m/s}^2$, $T = 27.6$ N.

5. A 2.5 kg box on a rough table is connected to a hanging 1.5 kg sphere by a light inextensible string over a smooth fixed pulley. Friction is k N. Released from rest, the sphere falls 0.8 m in 0.75 s.

a. Find acceleration, tension and k.

b. Explain how inextensibility was used.

5

$a = \dfrac{2(0.8)}{0.75^2} = \dfrac{128}{45} = 2.84\ \text{m/s}^2.$

$T = 1.5(g - a) = 10.4333\ldots = 10.4\ \text{N}.$

5

$k = T - 2.5a = 3.3222\ldots = 3.32\ \text{N}.$

Retain unrounded values during the calculation.

5

While the string is taut, its fixed length gives equal acceleration magnitudes for the box and sphere.